

often in LEDs is gallium-arsenide, though there are many variations on this basic compound, such as Al-GaAs or aluminium-gallium-indium-phosphide.

These compounds are members of the so-called III-V group of semiconductors, that is, compounds made of elements listed in columns III and V of the periodic table. By varying the precise composition of the semiconductor, the wavelength (and therefore the colour) of the emitted light can be changed.

LED emission is generally in the visible part of the spectrum (with wavelengths from 0.4 to 0.7 micrometre) or in the near infra-red (with wavelengths between 0.7 and 2.0 micrometres). The brightness of the light observed from an LED depends on the power emitted by the LED and on the relative sensitivity of the eye at the emitted wavelength. Maximum sensitivity occurs at 0.555 micrometre, which is in the yellow-orange and green region. The applied voltage in most LEDs is quite low, in the region of two volts. The current depends on the application and ranges from a few milliamperes to several-hundred milliamperes.

The term diode refers to the twin-terminal structure of an LED. In a flashlight, for example, a wire filament is connected to a battery through two terminals—one (the anode) bearing the negative electric charge and the other (the cathode) bearing the positive charge. In LEDs, as in other semiconductor devices such as transistors, the terminals are actually two semiconductor materials of different composition and electronic properties brought together to form a junction.

In one material (the negative, or n-type semiconductor), the charge carriers are electrons, and in the other (the positive, or p-type semiconductor), the charge carriers are holes created by the absence of electrons. Under the influence of an electric field (supplied by a battery, for instance, when the LED is

switched on), current can be made to flow across p-n junction, providing the electronic excitation that causes the material to luminesce.

In a typical LED structure, the clear epoxy dome serves as a structural element to hold the lead frame together, as a lens to focus the light and as a refractive-index match to permit more light to escape from the LED chip. The chip, typically $250 \times 250 \times 250$ micrometres in dimension, is mounted in a reflecting cup formed in the lead frame.

The p-n-type GaP:N layers represent nitrogen added to gallium-phosphide to give green emission, the p-n-type GaAsP:N layers represent nitrogen added to gallium-arsenide-phosphide to give orange and yellow emission, and the p-type GaP:Zn,O layer represents zinc and oxygen added to gallium-phosphide to give red emission.

Two further enhancements, developed in the 1990s, are LEDs based on aluminium-gallium-indium-phosphide, which emit light efficiently from green to red-orange, and also blue LEDs based on silicon-carbide or gallium-nitride. Blue LEDs can be combined on a cluster with other LEDs to give all colours, including white, for full-colour moving displays.

Producing light

If we try to run current the other way, with the p-type side connected to the negative end of the circuit and the n-type side connected to the positive end, current will not flow. The negative electrons in the n-type material are attracted to the positive electrode. The positive holes in the p-type material are attracted to the negative electrode. No current flows across the junction because the holes and the electrons are each moving in the wrong direction. The depletion zone increases. The interaction

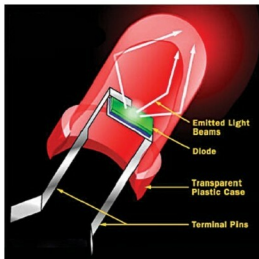


Fig. 4: Inside an LED

between electrons and holes in this setup has an interesting side effect—it generates light.

Light is a form of energy that can be released by an atom. It is made up of many small, particle-like packets that have energy and momentum but no mass. These particles, called photons, are the most basic units of light. Photons are released as a result of moving electrons. In an atom, electrons move in orbitals around the nucleus. Electrons in different orbitals have different amounts of energy. Generally speaking, electrons with greater energy move in orbitals farther away from the nucleus. For an electron to jump from a lower orbital to a higher orbital, something has to boost its energy level.

Conversely, an electron releases energy when it drops from a higher orbital to a lower one. This energy is released in the form of a photon. A greater energy-drop releases a higher-energy photon, which is characterised by a higher frequency.

In case of LEDs, free electrons moving across a diode can fall into empty holes from the p-type layer. This involves a drop from the conduction band to a lower orbital, so electrons release energy in the form of photons. This happens in any diode, but you can only see the photons when the diode is made of certain material.